

Section 26 Requirements

26.1 Current transformer requirements

The performance of a protection function will depend on the quality of the measured current signal. Saturation of the current transformers (CTs) will cause distortion of the current signals and can result in a failure to operate or cause unwanted operations of some functions. Consequently CT saturation can have an influence on both the dependability and the security of the protection. This protection IED has been designed to permit heavy CT saturation with maintained correct operation.

26.1.1 Current transformer basic classification and requirements

To guarantee correct operation, the current transformers (CTs) must be able to correctly reproduce the current for a minimum time before the CT will begin to saturate. To fulfill the requirement on a specified time to saturation the CTs must fulfill the requirements of a minimum secondary e.m.f. that is specified below.

CTs are specified according to many different classes and standards. In principle, there are three different types of protection CTs. These types are related to the design of the iron core and the presence of airgaps. Airgaps affects the properties of the remanent flux.

The following three different types of protection CTs have been specified:

- The High Remanence type with closed iron core and no specified limit of the remanent flux
- The Low Remanence type with small airgaps in the iron core and the remanent flux limit is specified to be maximum 10% of the saturation flux
- The Non Remanence type with big airgaps in the iron core and the remanent flux can be neglected

Even though no limit of the remanent flux is specified in the IEC standard for closed core CTs, it is a common opinion that the remanent flux is normally limited to maximum 75 - 80 % of the saturation flux.

Since approximately year 2000 some CT manufactures have introduced new core materials that gradually have increased the possible maximum levels of remanent flux even up to 95 % related to the hysteresis curve. Corresponding level of actual remanent flux is 90 % of the saturation flux (Ψ_{sat}). As the present CT standards have no limitation of the level of remanent flux, these CTs are also classified as for example, class TPX, P and PX according to IEC. The IEC TR 61869-100, Edition 1.0 2017-01, Instrument transformers – Guidance for application of current transformers in power system protection, is the first official document that highlighted this development. So far remanence factors of maximum 80% have been considered when CT requirements have been decided for Hitachi Energy IEDs. Even in the future this level of remanent flux probably will be the maximum level that will be considered when decided the CT requirements. If higher remanence levels should be considered, it should often lead to unrealistic CT sizes.

Thus, now there is a need to limit the acceptable level of remanent flux. To be able to guarantee the performance of protection IEDs, we need to introduce the following classification of CTs.

There are many different standards and a lot of classes but fundamentally there are four different types of CTs:

- Very High Remanence type CT
- High Remanence type CT
- Low Remanence type CT
- Non Remanence type CT

The Very High Remanence (VHR) type is a CT with closed iron core (for example, protection classes TPX, P, PX according to IEC, class C, K according to ANSI/IEEE) and with an iron core material (new material, typically new alloy based magnetic materials) that gives a remanent flux higher than 80 % of the saturation flux.

The High Remanence (HR) type is a CT with closed iron core (for example, protection classes TPX, P, PX according to IEC, class C, K according to ANSI/IEEE) but with an iron core material (traditional material) that gives a remanent flux that is limited to maximum 80 % of the saturation flux.

The Low Remanence (LR) type is a CT with small airgaps in the iron core (for example, TPY, PR, PXR according to IEC) and the remanent flux limit is specified to be maximum 10% of the saturation flux.

The Non Remanence (NR) type is a CT with big airgaps in the core (for example, TPZ according to IEC) and the remanent flux can be neglected.

It is also possible that different CT classes of HR and LR type may be mixed.

CT type VHR (using new material) should not be used for protection CT cores. This means that it is important to specify that the remanence factor must not exceed 80 % when ordering for example, class P, PX or TPX CTs. If CT manufacturers are using new core material and are not able to fulfill this requirement, the CTs shall be specified with small airgaps and therefore will be CTs of LR type (for example, class PR, TPY or PXR). Very high remanence level in a protection core CT can cause the following problems for protection IEDs:

1. Unwanted operation of differential (i.e. unit) protections for external faults
2. Unacceptably delayed or even missing operation of all types of protections (for example, distance, differential, overcurrent, etc.) which can result in losing protection selectivity in the network

No information is available about how frequent the use of the new iron core material is for protection CT cores, but it is known that some CT manufacturers are using the new material while other manufacturers continue to use the old traditional core material for protection CT cores. In a case where VHR type CTs have been already installed, the calculated values of E_{al} for HR type CTs, for which the formulas are given in this document, must be multiplied by factor two-and-a-half in order for VHR type CTs (i.e. with new material) to be used together with Hitachi Energy protection IEDs. However, this may result in unacceptably big CT cores, which can be difficult to manufacture and fit in available space.

Different standards and classes specify the saturation e.m.f. in different ways but it is possible to approximately compare values from different classes. The rated equivalent limiting secondary e.m.f. E_{al} according to the IEC 61869–2 standard is used to specify the CT requirements for the IED. The requirements are also specified according to other standards.

26.1.2 Conditions

The requirements are a result of investigations performed in our network simulator. The current transformer models are representative for current transformers of high remanence and low remanence type. The results may not always be valid for non remanence type CTs (TPZ).

The performances of the protection functions have been checked in the range from symmetrical to fully asymmetrical fault currents. Primary time constants of at least 120 ms have been considered at the tests. The current requirements below are thus applicable both for symmetrical and asymmetrical fault currents.

Depending on the protection function phase-to-earth, phase-to-phase and three-phase faults have been tested for different relevant fault positions for example, close in forward and reverse faults, zone 1 reach faults, internal and external faults. The dependability and security of the protection was verified by checking for example, time delays, unwanted operations, directionality, overreach and stability.

The remanence in the current transformer core can cause unwanted operations or minor additional time delays for some protection functions. As unwanted operations are not acceptable at all maximum remanence has been considered for fault cases critical for the security, for example, faults in reverse direction and external faults. Because of the almost negligible risk of additional time delays and the non-existent risk of failure to operate the remanence have not been considered for the dependability cases. The requirements below are therefore fully valid for all normal applications.

It is difficult to give general recommendations for additional margins for remanence to avoid the minor risk of an additional time delay. They depend on the performance and economy requirements. When current transformers of low remanence type (for example, TPY, PR) are used, normally no additional margin is needed. For current transformers of high remanence type (for example, P, PX, TPX) the small probability of fully asymmetrical faults, together with high remanence in the same direction as the flux generated by the fault, has to be kept in mind at the decision of an additional margin. Fully asymmetrical fault current will be achieved when the fault occurs at approximately zero voltage (0°). Investigations have shown that 95% of the faults in the network will occur when the voltage is between 40° and 90° . In addition fully asymmetrical fault current will not exist in all phases at the same time.

26.1.3 Fault current

The current transformer requirements are based on the maximum fault current for faults in different positions. Maximum fault current will occur for three-phase faults or single phase-to-earth faults. The current for a single phase-to-earth fault will exceed the current for a three-phase fault when the zero sequence impedance in the total fault loop is less than the positive sequence impedance.

When calculating the current transformer requirements, maximum fault current for the relevant fault position should be used and therefore both fault types have to be considered.

26.1.4 Secondary wire resistance and additional load

The voltage at the current transformer secondary terminals directly affects the current transformer saturation. This voltage is developed in a loop containing the secondary wires and the burden of all relays in the circuit. For earth faults the loop includes the phase and neutral wire, normally twice the resistance of the single secondary wire. For three-phase faults the neutral current is zero and it is just necessary to consider the resistance up to the point where the phase wires are connected to the common neutral wire. The most common practice is to use four wires secondary cables so it normally is sufficient to consider just a single secondary wire for the three-phase case.

The conclusion is that the loop resistance, twice the resistance of the single secondary wire, must be used in the calculation for phase-to-earth faults and the phase resistance, the resistance of a single secondary wire, may normally be used in the calculation for three-phase faults.

As the burden can be considerable different for three-phase faults and phase-to-earth faults it is important to consider both cases. Even in a case where the phase-to-earth fault current is smaller than the three-phase fault current the phase-to-earth fault can be dimensioning for the CT depending on the higher burden.

In isolated or high impedance earthed systems the phase-to-earth fault is not the dimensioning case. Therefore, the resistance of the single secondary wire can always be used in the calculation for this kind of power systems.

26.1.5 General current transformer requirements

The current transformer ratio is mainly selected based on power system data for example, maximum load and/or maximum fault current. It should be verified that the current to the protection is higher than the minimum operating value for all faults that are to be detected with the selected CT ratio. It should also be verified that the maximum possible fault current is within the limits of the IED.

The current error of the current transformer can limit the possibility to use a very sensitive setting of a sensitive residual overcurrent protection. If a very sensitive setting of this function will be used it is recommended that the current transformer should have an accuracy class which have an current error at rated primary current that is less than $\pm 1\%$ (for example, 5P). If current transformers with less accuracy are used it is advisable to check the actual unwanted residual current during the commissioning.

26.1.6 Rated equivalent secondary e.m.f. requirements

With regard to saturation of the current transformer all current transformers of high remanence and low remanence type that fulfill the requirements on the rated equivalent limiting secondary e.m.f. E_{al} below can be used. The characteristic of the non remanence type CT (TPZ) is not well defined as far as the phase angle error is concerned. If no explicit recommendation is given for a specific function we therefore recommend contacting Hitachi Energy to confirm that the non remanence type can be used.

The CT requirements for the different functions below are specified as a rated equivalent limiting secondary e.m.f. E_{al} according to the IEC 61869-2 standard. Requirements for CTs specified according to other classes and standards are given at the end of this section.

26.1.6.1 Transformer differential protection

The current transformers must have a rated equivalent limiting secondary e.m.f. E_{al} that is larger than the maximum of the required rated equivalent limiting secondary e.m.f. E_{alreq} below:

$$E_{al} \geq E_{alreq} = 30 \cdot I_{rt} \cdot \frac{I_{sr}}{I_{pr}} \cdot \left(R_{ct} + R_L + \frac{S_R}{I_r^2} \right)$$

(Equation 573)

$$E_{al} \geq E_{alreq} = 2 \cdot I_{tf} \cdot \frac{I_{sr}}{I_{pr}} \cdot \left(R_{ct} + R_L + \frac{S_R}{I_r^2} \right)$$

(Equation 574)

where:

I_{rt}	The rated primary current of the power transformer (A)
I_{tf}	Maximum primary fundamental frequency current that passes two main CTs and the power transformer (A)
I_{pr}	The rated primary CT current (A)
I_{sr}	The rated secondary CT current (A)
I_r	The rated current of the protection IED (A)
R_{ct}	The secondary resistance of the CT (W)
R_L	The resistance of the secondary wire and additional load (W). The loop resistance containing the phase and neutral wires must be used for faults in solidly earthed systems. The resistance of a single secondary wire should be used for faults in high impedance earthed systems.
S_R	The burden of an IED current input channel (VA). $S_R=0.020$ VA/channel for $I_r=1$ A and $S_R=0.150$ VA/channel for $I_r=5$ A

In substations with breaker-and-a-half or double-busbar double-breaker arrangement, the fault current may pass two main CTs for the transformer differential protection without passing the power transformer. In such cases and if both main CTs have equal ratios and magnetization characteristics the CTs must satisfy [Equation 573](#) and [Equation 575](#).

$$E_{al} \geq E_{alreq} = I_f \cdot \frac{I_{sr}}{I_{pr}} \cdot \left(R_{ct} + R_L + \frac{S_R}{I_r^2} \right)$$

(Equation 575)

where:

I_f Maximum primary fundamental frequency current that passes two main CTs without passing the power transformer (A)

26.1.6.2 Distance protection

The current transformers must have a rated equivalent limiting secondary e.m.f. E_{al} that is larger than the maximum of the required rated equivalent limiting secondary e.m.f. E_{alreq} below:

$$E_{al} \geq E_{alreq} = \frac{I_{kmax} \cdot I_{sr}}{I_{pr}} \cdot a \cdot \left(R_{ct} + R_L + \frac{S_R}{I_r^2} \right)$$

(Equation 576)

$$E_{al} \geq E_{alreq} = \frac{I_{kzone1} \cdot I_{sr}}{I_{pr}} \cdot k \cdot \left(R_{ct} + R_L + \frac{S_R}{I_r^2} \right)$$

(Equation 577)

where:

I_{kmax}	Maximum primary fundamental frequency current for close-in forward and reverse faults (A)
I_{kzone1}	Maximum primary fundamental frequency current for faults at the end of zone 1 reach (A)
I_{pr}	The rated primary CT current (A)
I_{sr}	The rated secondary CT current (A)
I_r	The rated current of the protection IED (A)
R_{ct}	The secondary resistance of the CT (W)
R_L	The resistance of the secondary wire and additional load (W). In solidly earthed systems the loop resistance containing the phase and neutral wires should be used for phase-to-earth faults and the resistance of the phase wire should be used for three-phase faults. In isolated or high impedance earthed systems the resistance of the single secondary wire can always be used.
S_R	The burden of an IED current input channel (VA). $S_R=0.020$ VA/channel for $I_r=1$ A and $S_R=0.150$ VA/channel for $I_r=5$ A
a	This factor depends on the design of the protection function and can be a function of the primary DC time constant of the close-in fault current.
k	This factor depends on the design of the protection function and can be a function of the primary DC time constant of the fault current for a fault at the set reach of zone 1.

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The a- and k-factors have the following values for the different types of distance function:

High speed distance: (ZMFPDIS and ZMFCPDIS)

Quadrilateral characteristic:

a = 1 for primary time constant Tp £ 400 ms

k = 3 for primary time constant Tp £ 200 ms

Mho characteristic:

a = 2 for primary time constant Tp £ 400 ms (For a = 1 the delay in operation due to saturation is still under 1.5 cycles)

k = 3 for primary time constant Tp £ 200 ms

Quadrilateral distance: (ZMQPDIS, ZMQAPDIS and ZMCPDIS, ZMCAPDIS and ZMMPDIS, ZMMPAPDIS)

a = 1 for primary time constant Tp £ 100 ms

a = 3 for primary time constant Tp > 100 and £ 400 ms

k = 4 for primary time constant Tp £ 50 ms

k = 5 for primary time constant Tp > 50 and £ 150 ms

Mho distance: (ZMHPDIS)

a = 1 for primary time constant Tp £ 100 ms

a = 3 for primary time constant Tp > 100 and £ 400 ms

k = 4 for primary time constant Tp £ 40 ms

k = 5 for primary time constant Tp > 40 and £ 150 ms

26.1.6.3 Breaker failure protection

The CTs must have a rated equivalent limiting secondary e.m.f. E_{al} that is larger than or equal to the required rated equivalent limiting secondary e.m.f. E_{alreq} below:

$$E_{a1} \geq E_{a1req} = 5 \cdot I_{op} \cdot \frac{I_{sr}}{I_{pr}} \cdot \left(R_{ct} + R_L + \frac{S_R}{I_r^2} \right)$$

(Equation 578)

where:

I_{op} The primary operate value (A)

I_{pr} The rated primary CT current (A)

I_{sr} The rated secondary CT current (A)

I_r The rated current of the protection IED (A)

R_{ct} The secondary resistance of the CT (W)

R_L The resistance of the secondary cable and additional load (W). The loop resistance containing the phase and neutral wires, must be used for faults in solidly earthed systems. The resistance of a single secondary wire should be used for faults in high impedance earthed systems.

S_R The burden of an IED current input channel (VA). $S_R=0.020$ VA/channel for $I_r=1$ A and $S_R=0.150$ VA/channel for $I_r=5$ A

26.1.6.4 Restricted earth fault protection (low impedance differential)

The requirements are specified separately for solidly earthed and impedance earthed transformers. For impedance earthed transformers, the requirements for the phase CTs are depending whether it is three individual CTs connected in parallel or it is a cable CT enclosing all three phases.

Neutral CTs and phase CTs for solidly earthed transformers

The neutral CT and the phase CTs must have a rated equivalent limiting secondary e.m.f. E_{al} that is larger than or equal to the maximum of the required rated equivalent limiting secondary e.m.f. E_{alreq} below:

$$E_{al} \geq E_{alreq} = 30 \cdot I_{rt} \cdot \frac{I_{sr}}{I_{pr}} \cdot \left(R_{ct} + R_L + \frac{S_R}{I_r^2} \right)$$

(Equation 579)

$$E_{al} \geq E_{alreq} = 2 \cdot I_{eff} \cdot \frac{I_{sr}}{I_{pr}} \cdot \left(R_{ct} + R_L + \frac{S_R}{I_r^2} \right)$$

(Equation 580)

Where:

I_{rt}	The rated primary current of the power transformer (A)
I_{eff}	Maximum primary fundamental frequency phase-to-earth fault current that passes the CTs and the power transformer neutral (A)
I_{pr}	The rated primary CT current (A)
I_{sr}	The rated secondary CT current (A)
I_r	The rated current of the protection IED (A)
R_{ct}	The secondary resistance of the CT (Ω)
R_L	The resistance of the secondary wire and additional load (Ω). The loop resistance containing the phase and neutral wires shall be used.
S_R	The burden of a REx670 current input channel (VA). $S_R=0.020$ VA / channel for $I_r = 1$ A and $S_R = 0.150$ VA / channel for $I_r = 5$ A

In substations with breaker-and-a-half or double-busbar double-breaker arrangement, the fault current may pass two main phase CTs for the restricted earth fault protection without passing the power transformer. In such cases and if both main CTs have equal ratios and magnetization characteristics the CTs must satisfy Requirement [Equation 579](#) and the Requirement [Equation 580](#):

$$E_{al} \geq E_{alreq} = I_{ef} \cdot \frac{I_{sr}}{I_{pr}} \cdot \left(R_{ct} + R_L + \frac{S_R}{I_r^2} \right)$$

(Equation 581)

Where:

I_{ef}	Maximum primary fundamental frequency phase-to-earth fault current that passes two main CTs without passing the power transformer neutral (A)
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Neutral CTs and phase CTs for impedance earthed transformers

The neutral CT and phase CTs must have a rated equivalent limiting secondary e.m.f. E_{al} that is larger than or equal to the required rated equivalent limiting secondary e.m.f. E_{alreq} below:

$$E_{al} \geq E_{alreq} = 3 \cdot I_{eff} \cdot \frac{I_{sr}}{I_{pr}} \cdot \left(R_{ct} + R_L + \frac{S_R}{I_r^2} \right)$$

(Equation 582)

Where:

I_{eff}	Maximum primary fundamental frequency phase-to-earth fault current that passes the CTs and the power transformer neutral (A)
I_{pr}	The rated primary CT current (A)
I_{sr}	The rated secondary CT current (A)
I_r	The rated current of the protection IED (A)
R_{ct}	The secondary resistance of the CT (Ω)
R_L	The resistance of the secondary wire and additional load (Ω). The loop resistance containing the phase and neutral wires shall be used.
S_R	The burden of a REx670 current input channel (VA). $S_R = 0.020$ VA / channel for $I_r = 1$ A and $S_R = 0.150$ VA / channel for $I_r = 5$ A

In case of three individual CTs connected in parallel (Holmgren connection) on the phase side the following additional requirements must also be fulfilled.

The three individual phase CTs must have a rated equivalent limiting secondary e.m.f. E_{al} that is larger than or equal to the maximum of the required rated equivalent limiting secondary e.m.f. E_{alreq} below:

$$E_{al} \geq E_{alreq} = 2 \cdot I_{tf} \cdot \frac{I_{sr}}{I_{pr}} \cdot \left(R_{ct} + R_{Lsw} + \frac{S_R}{I_r^2} \right)$$

(Equation 583)

Where:

I_{tf} Maximum primary fundamental frequency three-phase fault current that passes the CTs and the power transformer (A).

R_{Lsw} The resistance of the single secondary wire and additional load (Ω).

In impedance earthed systems the phase-to-earth fault currents often are relatively small and the requirements might result in small CTs. However, in applications where the zero sequence current from the phase side of the transformer is a summation of currents from more than one CT (cable CTs or groups of individual CTs in Holmgren connection) for example, in substations with breaker-and-a-half or double-busbar double-breaker arrangement or if the transformer has a T-connection to different busbars, there is a risk that the CTs can be exposed for higher fault currents than the considered phase-to-earth fault currents above. Examples of such cases can be cross-country faults or phase-to-phase faults with high fault currents and unsymmetrical distribution of the phase currents between the CTs. The zero sequence fault current level can differ much and is often difficult to calculate or estimate for different cases. To cover these cases, with summation of zero sequence currents from more than one CT, the phase side CTs must fulfill the Requirement [Equation 584](#) below:

$$E_{al} \geq E_{alreq} = I_f \cdot \frac{I_{sr}}{I_{pr}} \cdot \left(R_{ct} + R_L + \frac{S_R}{I_r^2} \right)$$

(Equation 584)

Where:

I_f Maximum primary fundamental frequency three-phase fault current that passes the CTs (A)

R_L The resistance of the secondary wire and additional load (Ω). The loop resistance containing the phase and neutral wires shall be used.

26.1.7 Current transformer requirements for CTs according to other standards

All kinds of conventional magnetic core CTs are possible to use with the IEDs if they fulfill the requirements corresponding to the above specified expressed as the rated equivalent limiting secondary e.m.f. E_{al} according to the IEC 61869-2 standard. From different standards and available data for relaying applications it is possible to approximately calculate a secondary e.m.f. of the CT comparable with E_{al} . By comparing this with the required rated equivalent limiting secondary e.m.f. E_{alreq} it is possible to judge if the CT fulfills the requirements. The requirements according to some other standards are specified below.

26.1.7.1 Current transformers according to IEC 61869-2, class P, PR

A CT according to IEC 61869-2 is specified by the secondary limiting e.m.f. E_{ALF} . The value of the E_{ALF} is approximately equal to the corresponding E_{al} . Therefore, the CTs according to class P and PR must have a secondary limiting e.m.f. E_{ALF} that fulfills the following:

$$E_{ALF} > \max E_{alreq}$$

(Equation 585)

26.1.7.2 Current transformers according to IEC 61869-2, class PX, PXR (and old IEC 60044-6, class TPS and old British Standard, class X)

CTs according to these classes are specified approximately in the same way by a rated knee point e.m.f. E_{knee} (E_k for class PX and PXR, E_{kneeBS} for class X and the limiting secondary voltage U_{al} for TPS). The value of the E_{knee} is lower than the corresponding E_{al} according to IEC 61869-2. It is not possible to give a general relation between the E_{knee} and the E_{al} but normally the E_{knee} is approximately 80 % of the E_{al} . Therefore, the CTs according to class PX, PXR, X and TPS must have a rated knee point e.m.f. E_{knee} that fulfills the following:

$$E_{knee} \approx E_k \approx E_{kneeBS} \approx U_{al} > 0.8 \cdot (\text{maximum of } E_{alreq})$$

(Equation 586)

26.1.7.3 Current transformers according to ANSI/IEEE

Current transformers according to ANSI/IEEE are partly specified in different ways. A rated secondary terminal voltage U_{ANSI} is specified for a CT of class C. U_{ANSI} is the secondary terminal voltage the CT will deliver to a standard burden at 20 times rated secondary current without exceeding 10 % ratio correction. There are a number of standardized U_{ANSI} values for example, U_{ANSI} is 400 V for a C400 CT. A corresponding rated equivalent limiting secondary e.m.f. E_{alANSI} can be estimated as follows:

$$E_{alANSI} = |20 \cdot I_{sr} \cdot R_{ct} + U_{ANSI}| = |20 \cdot I_{sr} \cdot R_{ct} + 20 \cdot I_{sr} \cdot Z_{bANSI}|$$

(Equation 587)

where:

Z_{bANSI} The impedance (that is, with a complex quantity) of the standard ANSI burden for the specific C class (W)

U_{ANSI} The secondary terminal voltage for the specific C class (V)

The CTs according to class C must have a calculated rated equivalent limiting secondary e.m.f. E_{alANSI} that fulfills the following:

$$E_{alANSI} > \text{maximum of } E_{alreq}$$

(Equation 588)

A CT according to ANSI/IEEE is also specified by the knee point voltage $U_{kneeANSI}$ that is graphically defined from an excitation curve. The knee point voltage $U_{kneeANSI}$ normally has a lower value than the knee-point e.m.f. according to IEC and BS. $U_{kneeANSI}$ can approximately be estimated to 75 % of the corresponding E_{al} according to IEC 61869-2. Therefore, the CTs according to ANSI/IEEE must have a knee point voltage $U_{kneeANSI}$ that fulfills the following:

$$U_{kneeANSI} > 0.75 \cdot (\text{maximum of } E_{alreq})$$

(Equation 589)

26.2 Voltage transformer requirements

The performance of a protection function will depend on the quality of the measured input signal. Transients caused by capacitive voltage transformers (CVTs) can affect some protection functions.

Magnetic or capacitive voltage transformers can be used.